Terms of Reference

Working Group 2: Actionable Cryosphere Projections

The Decade of Action for Cryospheric Sciences (2025-2034)

Background

On August 13, 2024, the United Nations General Assembly adopted Resolution 78/321, proclaiming 2025 to 2034 as the Decade of Action for Cryospheric Sciences. The resolution aims to accelerate efforts to address the urgent challenges posed by the melting cryosphere through enhanced scientific research, monitoring, and international cooperation. The resolution highlights the need to understand the cryosphere's role in the Earth's climate system, freshwater resources, ecosystems, and human social, political, and economic activities. It also calls for raising awareness of cryospheric loss and encourages the sharing of best practices to build resilience against related risks. Member States, UN bodies, and stakeholders are invited to support these activities. UNESCO is tasked with leading implementation of the Decade, with other UN organizations and partners, using voluntary contributions to develop programmes.

The mission in a nutshell: *WG2 connects scientific capability to decisions. It begins with the questions and decisions faced by affected communities, Indigenous Peoples, policymakers, practitioners, and other actors; identifies which modelling and projection capabilities can reliably inform those decisions; makes assumptions, uncertainties, and limitations transparent; and develops inclusive tools, use cases and coordination mechanisms through which projections can be surfaced, interpreted, actioned, and tracked.*

During the first year of the Decade, 2025, an Ad Hoc Committee was convened by UNESCO to form the structures necessary to implement the Decade. These are summarised in a **Synthesis Document**, which outlines the vision for the Decade. As a result of this process, the committee identified five Overarching Challenges (OCs) for the Decade:

OC1. Global Cryosphere Monitoring
OC2. Actionable Cryosphere Projections
OC3. Cryospheric Risk Management Solutions
OC4. Adaptation to Cryosphere Loss
OC5. Amplify the Impact of Cryospheric Science

Within the Decade, Working Groups aligned with each OC serve as the *main engines of delivery*. The role of Working Group 2 is to achieve the Main Goals (MGs) of **OC2, Actionable Cryosphere Projections**:

MG2.1: Cryosphere Modelling for Decision Making
MG2.2: Inclusive Cryosphere Prediction
MG2.3: Creation of a Decision and Outreach Toolkit

The Decade is split into three triennia. Each triennium has a primary Main Goal, marked by a triennial summit. All Main Goals remain active, in the scope, throughout the Decade and inform one another iteratively. It is anticipated that appointment to each working group will be on an initial three-year basis, with membership rotating as focus shifts towards subsequent MGs.

By the end of the Decade, this Working Group will have established a coordinated and inclusive framework that strengthens the societal value of cryosphere modelling: improves transparency and

accessibility of model outputs as well as interoperability and interpretability; and enables existing scientific capabilities to meaningfully inform concrete decisions, policy, planning, investment, and adaptation across relevant spatial and temporal scales.

Throughout the ten-year initiative, documents, questions, priorities and projections will evolve. For the sake of transparency and legitimacy, major questions, assumptions, outputs, and decisions shall be versioned and accompanied by a concise record of their rationale and revision history. Earlier versions shall remain accessible when technically possible, so that the evolution of WG2's understanding and priorities can be traced.

Definition (Actionable cryosphere projection). For the purposes of WG2, an "actionable cryosphere projection" is a scientifically grounded and qualified statement about possible attainable future cryosphere states. The statement includes the anticipated consequences of attainment and is developed in relation to a clearly defined pathway, timescale, and uncertainty context. In practice, actionability requires both scientific reliability and an explicit, transparent pathway connecting questions, projections, interpretation, actors, and decisions.

Definition (Consensus, workable definition). For the purposes of WG2, consensus means the absence of a reasoned and sustained objection after adequate consultation; it does not require uniform unanimity or the active approval of every member. Where "uniform consensus" cannot be reached, the differing positions shall be documented, the leadership may refer the matter to the UNESCO IHP Decade Secretariat or Strategic Management Committee, as appropriate.

Definition (Decision). A committed choice among available alternatives that changes future actions, allocation of resources, or planning.

Remark (Quorum, tentative). Leadership meetings may proceed when the Lead or a delegated Associate Lead, at least one further Lead or Associate Lead, the UNESCO Project Officer, and/or the Secretariat are represented. Members unable to attend shall be given a reasonable opportunity to contribute asynchronously.

Terminology. WG2 uses "modelling" for the scientific (conceptual and computational) representation of cryosphere processes; "projection" for conditional statements about possible future states under specified assumptions or scenarios; and "prediction" for time-bound anticipatory statements whose interpretation depends on initial conditions, available knowledge, and uncertainty.

Initial Objectives

Rapid changes in the cryosphere are reshaping global and regional climate systems, water resources, ecosystems, hazards, and socio-economic conditions. While substantial cryosphere modelling capacity already exists globally, efforts remain fragmented across disciplines, regions, and institutions, with limited integration of human systems, policy needs, Indigenous, local, as well as scientific knowledge systems. Modelling infrastructure and expertise are also unevenly distributed, constraining equitable participation and influencing research priorities.

OC2 of the Decade responds to these challenges by repositioning cryosphere modelling from model development alone towards coordination, synthesis, and relevance to decisions. The focus of the *first three years is on MG2.1 Cryosphere Modelling for Decision Making.* The first task is to review the state-of-the-art of modelling capabilities, focussing on the capacity to inform policy and decision making. This review will focus on:

- What can current models reliably answer for questions relevant to policymaking?
- What can current models reliably answer for questions relevant to Indigenous Peoples?
- What can current models reliably answer for questions relevant to affected communities?

- What are the key uncertainties and process gaps that limit actionable insights?
- Which geographical and thematic regions are underrepresented in current model efforts?

During the first triennium, MG2.1 constitutes the principal focus; preparatory work under MG2.2 and MG2.3 informs reviewing and testing under real-world scenarios and use cases.

The work of Working Group 2 over the first three years will be presented and discussed at the first *Triennial Summit of the Decade of Action on Cryosphere Sciences*.

Wider Mandate and Deliverables

What decisions, questions, timescales, and forms of evidence matter to affected actors, policymakers, Indigenous Peoples, local communities, and practitioners?

Remark (Flow at the boundaries of WG2). WG2 receives observational capabilities and data needs from WG1; develops projections and interpretable pathways; interacts with WG3 to translate projections into hazard and risk assessments; supports WG4 in applying projections to adaptation choices; and collaborates with WG5 on communication, education and societal uptake. Cross-group questions and outputs should be jointly scoped where mandates overlap.

As one of the Decade's five thematic Working Groups, the Working Group 2 is mandated to:

- Act as the primary delivery mechanism for OC2, reporting to and coordinated by the Strategic Management Committee;
- In the first three years (2026-2029), lead implementation of MG2.1 and support transition to MG2.2;
- Encourage model integration and application to answer urgent questions on the impacts of cryospheric decline;
- Coordinate with other Decade Working Groups, particularly those addressing observations (OC1) and hazards and impacts (OC3), to ensure coherence across the Decade;
- Engage with existing international, regional, and national modelling initiatives to ensure complementarity and avoid duplication.
- Integrate stakeholder needs, Indigenous and local knowledge, stakeholder needs and co-design into cryosphere modelling and prediction frameworks.
- Establish transparent mechanisms to gather, refine, prioritise and revisit decision-relevant questions from scientific, policy, Indigenous, local communities, practitioners and funding actors;
- Maintain a living overview of relevant modelling initiatives, capabilities, questions, use cases and planned contributions across the Decade, supporting coordination and avoiding duplication.
- Translate identified priorities and gaps into scientifically solid, ethically grounded and investment-ready project concepts, in coordination with the Finance Dialogue and relevant partners.

Throughout the decade, this Working Group will report biannually to the Strategic Management Committee, on progress towards the following deliverables:

1. A periodically updated review and gap analysis of cryosphere modelling capabilities, relevant to policy, Indigenous, and local community decision needs, including guidance on strengths, limitations, applicable spatial and temporal scales, uncertainties, and gaps.
2. Creation of a cryosphere modelling and decision forum, linking existing modelling communities, policymakers, funders, Indigenous representatives, and other stakeholders, to

formulate questions, interpret projections, review priorities, and promote the co-design of research.

3. Best-practice guidelines for policy-relevant and Indigenous-relevant cryosphere modelling.
4. An operational Decision and Outreach Toolkit that supports capacity building for policymakers, practitioners, and communities.
5. Use cases demonstrating how modelling and projections inform real decisions, participating actors, assumptions, uncertainties, alternatives, outcomes, and learning.
6. A progressive implementation and review framework for FAIR (Findable, Accessible, Interoperable, Reusable), CARE (Collective Benefit, Authority to Control, Responsibility, Ethics), and TRUST (Transparency, Responsibility, User Focus, Sustainability, Technology) principles within cryosphere modelling, data and prediction outputs;
7. A living, versioned coordination map of decision-relevant questions, modelling capabilities, active initiatives, planned contributions and identified knowledge/action gaps across the Decade.

Structure

a) Open and inclusive membership

WG2 is open to all interested experts and stakeholders contributing to the objectives of the Decade. General membership shall remain open to all applicants meeting the criteria of the Decade, with no fixed threshold or numerical cap, and all members shall be entitled to participate in the discussions and activities of the Group.

General members can participate in WG2 discussions, consultations and workstreams. The core group (see below) guarantees continuity, prepares recommendations, and coordinates deliverables. Endorsement (signature) of WG2 outputs and positions rests with the leadership, following consultation with the broader membership and in accordance with the consensus procedure described below.

b) Leadership composition

The leadership of WG2 comprises:

- a Lead, appointed for an initial three-year term;
- two Associate Leads, appointed for an initial three-year term, jointly with the Lead, on the basis of complementarity of expertise, thematic coverage and geographical balance;
- a Secretariat, hosted by an institution with the mandate, capacity, and networks to support the scientific coordination of the Working Group.

These four positions are nominated by the Ad Hoc Strategic Management Committee on the proposal of the UNESCO IHP Secretariat, following the open call for expressions of interest and considering scientific excellence, geographical and gender balance, interdisciplinary expertise, and diversity of background and generation.

c) Core group

A flexible core group shall be established to support coordination and continuity. The size and composition of the core group shall be determined by the Lead, Associate Leads and Secretariat, in consultation with the broader WG2 and the UNESCO IHP Secretariat, on the basis of thematic needs and the level of expressed interest. The core group shall as a guiding principle include:

- a transdisciplinary mix of scientific, policy, practitioner and knowledge-holder profiles;

- geographical balance, with deliberate inclusion of expertise from polar regions, high-mountain regions, snow and permafrost zones, as well as downstream and low-lying regions affected by cryosphere loss;
- dedicated seats for representatives of Indigenous peoples and local communities, in articulation with the Indigenous Advisory Committee of the Decade;
- dedicated seats for early-career scientists, as part of the next-generation capacity ambition of the Decade;
- expertise spanning cryosphere modelling (glaciers, snow, permafrost, sea ice, polar and high-mountain systems) together with hydrology and water resources, oceans, ecosystems, socio-economic systems and hazards, and including policymakers, funders, practitioners and representatives of Global South institutions;

Participation in the Working Group is voluntary and non-remunerated.

The composition of the core-group is documented against transparent criteria covering expertise, thematic and geographical representation, career stage, knowledge systems, capacity to contribute, and identified gaps. Membership may evolve as workstreams and needs change. Calls for missing expertise or representation may be issued whenever gaps are identified.

d) Roles and responsibilities

The responsibilities of the leadership positions are defined as follows:

- **The Lead** is responsible for the overall direction, scientific quality and delivery of WG2; represents the Group at the Strategic Management Committee and externally; chairs meetings; ensures alignment with the Synthesis Document, with the Decade's Overarching Challenges and Main Goals, and with the Triennial reporting cycle.
- **The two Associate Leads** share thematic, geographical, and outreach responsibilities; act jointly with the Lead on substantive decisions; ensure thematic completeness across the breadth of cryosphere modelling and prediction; deputise for the Lead when required; and lead specific sub-streams of work as agreed within the Group.
- **The Secretariat** provides operational, interface, and institutional support to the Group; convenes meetings and maintains documentation, decisions, action registers and versioned outputs; supports transparent consultation and participation; curates an overview of questions, initiatives, workstreams and milestones; promotes alignment and effective connection with the host organisation's networks and existing programmes; and liaises with the UNESCO IHP Decade Secretariat and with the Secretariats of the other Working Groups, the Finance Dialogue and the Alliance for the Cryosphere, to foster cross-cutting coordination.

WG2 may establish time-bound workstreams around defined questions, deliverables, regions, or use cases. Each workstream (sub-stream) shall have a clear purpose, lead, membership, expected output, timeline, and relationship to the wider WG2 mandate. Workstreams may include external contributors where relevant.

Funding relationships and potential conflicts of interest shall be declared and managed transparently.

e) Term, rotation and continuity

Leadership and core-group positions are appointed for an initial three-year term, aligned with the three triennia of the Decade and the Triennial Cryosphere Summits. A transitional overlap of up to six months between outgoing and incoming officers shall be maintained to preserve continuity. Mid-term review by the Strategic Management Committee may lead to adjustments where required.

Modus Operandi

- The Working Group *shall meet* primarily online (tentatively 2 hours each month), with a frequency sufficient to maintain momentum, complemented by periodic in-person meetings aligned with major international events where feasible and provided funding resources are made available.
- Each meeting shall conclude with a concise record of decisions, responsible persons, dependencies and target dates.
- WG2 *shall work* through a combination of synchronous meetings and asynchronous work, consultation, and exchange. Drafts, questions, decisions and action items shall remain accessible for a defined review period, allowing members across time zones and with differing availability to contribute.
- The Working Group shall hold at least one joint meeting per year with the other Decade Working Groups (OC1-OC5) to foster synergy, share progress on cross-cutting issues, align activities across Overarching Challenges, and avoid duplication of effort.
- Progress and outputs of the Working Group shall be reported biannually by the Core group to the Strategic Management Committee and showcased at the Triennial Cryosphere Summits.
- Draft agendas and documents shall be circulated in advance of each meeting. Decisions of the Core Group and the Working Group shall be made by consensus (cf. Definitions).
- Members of the Management Committee shall be expected to consult and engage with their respective networks and constituencies, and to facilitate broad participation in the activities of the Working Groups. The Lead of the Working Group serves as a member of the Strategic Management Committee and maintains regular contact with the UNESCO IHP Decade Secretariat.
- In line with the Synthesis Document, WG2 shall not endorse, fund, promote or otherwise legitimise geoengineering research or deployment, recognising the scientifically uncertain and potentially harmful nature of such interventions. Activities shall integrate Indigenous and local knowledge ethically, with consent, recognition and benefit-sharing, and shall comply with UNESCO rules and procedures.
- WG2 shall actively articulate its work with the International Polar Year (IPY 2032-2033), the UN Decade of Ocean Science for Sustainable Development (2021-2030), the International Decade of Sciences for Sustainable Development (2024-2033), and other relevant initiatives, to maximise synergies, and avoid duplication.

Website and further details

Proceedings of the Decade, consultation summaries and deliverables will be published on the Decade's official website: www.un-cryosphere.org